

V4.2 Toy Model 2E

Interference Recovery and Erasure Constraint

Rev A - Exploratory Quantum-Side Compatibility Report

Date: 2026.05.25 | Continuation after Toy Model 2D

Status: Toy Model 2E tests quantum-eraser compatibility. It shows that interference can be recovered only in conditioned eraser branches, while the unconditioned local state remains unchanged. It is not retrocausality and not time travel.

1. Purpose

Toy Model 2C showed decoherence as loss of accessible interference. Toy Model 2D showed local observer update as conditioning on a record path. Toy Model 2E now tests the reverse constraint: if which-path environment records are erased before they become an irreversible observer record, can interference become visible again?

The required answer from standard quantum mechanics is precise: interference can be recovered only after conditioning on the eraser outcome. Without conditioning, the local observer sees no change and no faster-than-light signal.

2. Starting state with which-path records

The system-environment state is:

$$|\Psi_{SE}\rangle = \alpha |0\rangle|E_0\rangle + \beta |1\rangle|E_1\rangle$$

Here $|E_0\rangle$ and $|E_1\rangle$ are distinguishable environment records. If they are orthogonal, the local system state loses visible interference:

$$\rho_S = \begin{bmatrix} |\alpha|^2 & 0 \\ 0 & |\beta|^2 \end{bmatrix}$$

Plain meaning: which-path information exists in the environment, so the observer cannot see interference in the system alone.

3. Eraser basis

Instead of reading the environment in the which-path basis $|E_0\rangle, |E_1\rangle$, the eraser reads it in a mixed basis:

$$\begin{aligned} |E_+\rangle &= (|E_0\rangle + |E_1\rangle) / \sqrt{2} \\ |E_-\rangle &= (|E_0\rangle - |E_1\rangle) / \sqrt{2} \end{aligned}$$

This eraser basis does not preserve which-path information. It recombines the relational phase information into conditional branches.

4. Conditional interference recovery

If the eraser outcome is +, the conditioned system state becomes:

$$|\psi_+\rangle = \alpha |0\rangle + \beta |1\rangle$$

If the eraser outcome is -, the conditioned system state becomes:

$$|\psi_-\rangle = \alpha |0\rangle - \beta |1\rangle$$

Both conditioned branches recover off-diagonal coherence. They are complementary interference patterns.

Important: the branches must be sorted by eraser outcome. Without sorting, the + and - interference terms cancel.

5. Unconditioned no-signalling constraint

If the eraser outcome is not known locally, the observer must average over both eraser outcomes:

$$\rho_{\text{unread}} = P(+)\rho_+ + P(-)\rho_-$$

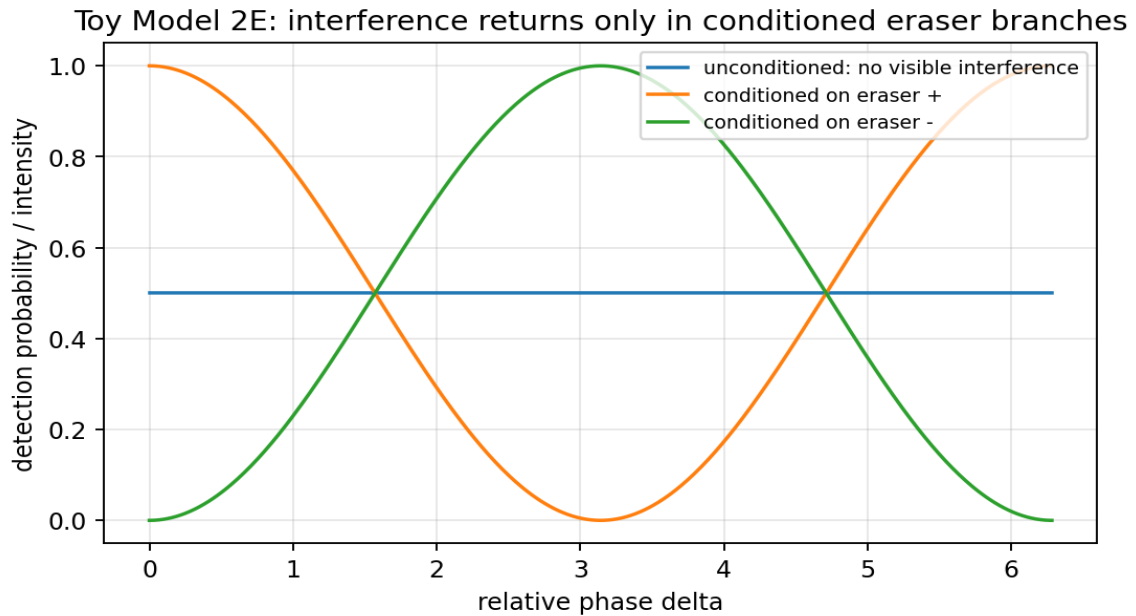
For the eraser basis above, this average reconstructs the same decohered state:

$$\rho_{\text{unread}} = \begin{bmatrix} |\alpha|^2 & 0 \\ 0 & |\beta|^2 \end{bmatrix}$$

Therefore, erasure cannot be used to send an instant signal. Interference recovery appears only after classical comparison with the eraser record.

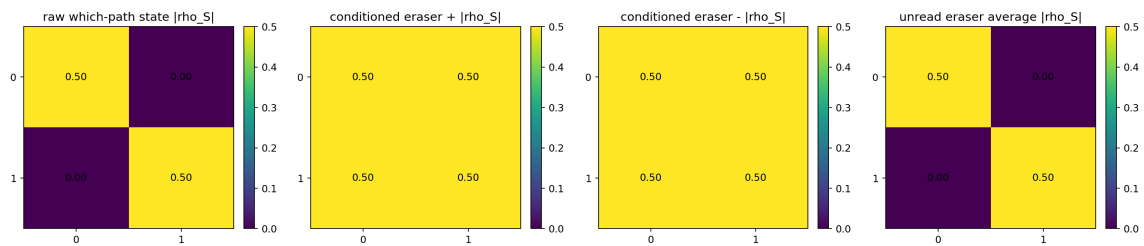
6. Numerical demonstration

For a balanced two-path state, the conditioned eraser branches show complementary interference patterns. The unconditioned pattern stays flat.



7. Density matrix snapshots

The raw which-path state has no off-diagonal coherence. The conditioned eraser branches recover off-diagonal coherence. The unread average cancels coherence again.



8. Numerical constraint checks

Check	Result
Random states tested	10000
Max global norm error	5.551e-16
Max raw trace error	3.338e-16
Max conditioned trace error	4.450e-16
Max eraser probability-sum error	5.551e-16
Max unconditioned reconstruction error	6.664e-16
Max plus-branch coherence restoration error	5.551e-16
Max minus-branch coherence restoration error	5.551e-16
Max no-signalling/unread-state error	6.664e-16

9. Interpretation inside the 6D Projection Framework

Toy Model 2E supports the following limited interpretation:

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which-path environment record
-> phase relation inaccessible to local 1T observer
-> eraser basis recombines record alternatives
-> conditioned branches recover interference
-> unread average remains unchanged

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In 6D language, the relational phase information was not destroyed. It was inaccessible because it was dispersed into distinguishable records. Erasure works only when those records are recombined before becoming an irreversible macroscopic observer record.

10. What Toy Model 2E achieves

- It shows how suppressed interference can reappear in conditioned eraser branches.

- It confirms that unread eraser outcomes do not change the local reduced state.
- It preserves probability normalization and global unitary compatibility.
- It clarifies that quantum erasure is not retrocausality.
- It strengthens the interpretation of decoherence as inaccessible relational phase information, not destroyed information.

11. What Toy Model 2E does not achieve

- It does not prove physical 3T existence.
- It does not solve the measurement problem.
- It does not derive the Born rule.
- It does not allow time travel or backwards causation.
- It does not create usable faster-than-light signalling.
- It does not yet model macroscopic irreversibility quantitatively.

12. Close-out statement for 2E

Toy Model 2E successfully formalizes interference recovery under an erasure constraint. A which-path environment record suppresses local interference, but measuring the environment in an eraser basis restores interference in conditioned subensembles. The unread average remains unchanged, preserving no-signalling. Within the reciprocal projection framework, this supports the view that decoherence makes phase information inaccessible to the 1T observer rather than destroying it globally. This is a compatibility result, not a derivation of Quantum Mechanics or evidence of retrocausality.

13. Recommended next chapter

The next logical chapter is Toy Model 2F: Macroscopic Irreversibility and Record Stability Threshold.

1. Model many environment records rather than one record qubit.
2. Show how interference recovery becomes exponentially impractical as the number of records grows.
3. Define a record-stability threshold for the 1T observer.
4. Separate in-principle reversibility from practical irreversibility.
5. Keep no-signalling, probability normalization, and global unitary compatibility as hard guardrails.

14. Rebind prompt for continuation

We are resuming after Toy Model 2E.

2E tested interference recovery and erasure constraint.

Starting state:

$$|\Psi_{SE}\rangle = \alpha|0\rangle|E_0\rangle + \beta|1\rangle|E_1\rangle$$

If $|E_0\rangle$ and $|E_1\rangle$ are distinguishable, local interference is suppressed:

$$\rho_S = \begin{bmatrix} |\alpha|^2 & 0 \\ 0 & |\beta|^2 \end{bmatrix}$$

Eraser basis:

$$|E_+\rangle = (|E_0\rangle + |E_1\rangle) / \sqrt{2}$$

$$|E_-\rangle = (|E_0\rangle - |E_1\rangle) / \sqrt{2}$$

Conditioned branches:

$$|\psi_+\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$|\psi_-\rangle = \alpha|0\rangle - \beta|1\rangle$$

Result:

Interference returns only in conditioned eraser branches.

The unread average remains the same decohered state, preserving no-signalling.

Guardrail:

No retrocausality, no time travel, no faster-than-light signalling.

This is compatibility with quantum eraser logic, not proof of physical 3T existence.

Next recommended chapter:

Toy Model 2F - Macroscopic Irreversibility and Record Stability Threshold.